Prediction Market Optimization

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Problem Statement - Why Prediction Market?

- Prediction markets are a real-money, real-time setting where beliefs update quickly.
- Categories (sports, elections, geopolitics, macro, etc.) create a natural cross-domain diversification problem.
- Core proposal idea: cap domain exposure while pursuing risk-adjusted returns with an online optimization algorithm.
- **Week 9 Changes:** Week 8 showed that a naive setup (ratio objective, heavy uniform mixing, huge universes) can look “optimized” while still behaving like equal weight. The project story is now **methodological iteration**: smoother training objective, scaled data frequency, smaller tradable set, Bayesian tuning, and honest holdout + attribution.

Process Overview

1. Ingest active events
2. Flatten + domain tagging
   - Convert event->market nested JSON to market rows, map tags to category/domain, and keep YES token IDs for binary interpretation.
3. Category-balanced market selection
   - Prioritize high-liquidity categories, apply per-category caps
4. Fetch token histories
5. Build cashed Week 8 dataset
6. Baseline run
   - Build price matrix, compute returns, allocate equal weight by domain, and save baseline metrics/time series.
7. Constrained run + running
8. Holdout evaluation + diagnostics

Core Math Formulas (1/2) - Definitions

- $r_t \in R^N$: vector of token returns at step t
- $w_t \in \Delta^N$: portfolio weights on simplex, $\sum_i w_{t,i} = 1$ and $w_{t,i} \geq 0$
- D: set of event domains/categories
- $S_d(w) = \sum_{i \in d} w_i$: total capital allocated to domain d
- L_d: domain limit for domain d

Core Math Formulas (2/2) - Concepts

$$R_t = \tilde{w}t^\top r_t, \qquad V_t = \prod s = 1^t(1 + R_s), \qquad \mathrm{DD}t = \frac{V_t}{\max s \le tV_s} - 1$$

$$\mathrm{Sortino}(R) = \frac{E[R]}{\sqrt{E[(\min(R,0))^2]} + \varepsilon}$$

$$\tilde{w}_t = (1-\alpha)\mathrm{softmax}(z_t) + \alpha$$

$$P_{\mathrm{domain}}(\tilde{w}) = \sum_{d \in \mathcal{D}} \max(0, S_d(\tilde{w}) - L_d)^2, \qquad P_{\mathrm{conc}}(\tilde{w}) = \sum_i \max(0, w_i - w_{\max})^2$$

$$P_{\mathrm{cov}}(\tilde{w}) = \tilde{w}^\top \Sigma \tilde{w} (\text{sample cov on the rolling window, with diagonal shrinkage})$$

$$B_{\mathrm{ent}}(\tilde{w}) = -\sum_i w_i \log(w_i + \varepsilon)$$

$$J_{\mathrm{md}}(R) = E[R] - \alpha_v \mathrm{Var}(R) - \beta_d E\big[\max(0, -R)^2\big]$$

Full per-step training objective (maximize; code minimizes $-$`obj`):

$$\max_{\tilde{w}} \Big[J_{\mathrm{md}}(R_{\mathrm{window}}(\tilde{w})) -_{\mathrm{domain}} (\tilde{w}) - \lambda_c P_{\mathrm{conc}}(\tilde{w}) - \lambda_\Sigma P_{\mathrm{cov}}(\tilde{w}) + \lambda_e B_{\mathrm{ent}}(\tilde{w}) \Big]$$

Evaluation Design

- Walk-forward tuning: Hyperparameters selected using rolling in-sample windows to ***optimize mean–downside objective on rolling windows***.
- Holdout testing: Final performance evaluated strictly out-of-sample on unseen data.
- Baseline comparison: Equal-weight portfolio used as benchmark for return, volatility, and drawdown.
- Metrics tracked: Cumulative return, max drawdown, volatility, rolling mean return, and return distribution

Biggest Changes

Topic	Week 8 flash guide	Week 9 (current)
Training target	Sortino ratio in-window	**Mean–downside** surrogate; Sortino for **reporting**
Uniform mix	Often high (e.g. 0.88 in old snapshot)	Search over **low** floors, including 0 (`uniform_mixes` in pipeline)
Inner optimization	Single-step narrative	`**steps_per_window=5` **, **Adam**
Evaluation	Frozen vs online was a Week 8 issue	Default **online** holdout
Universe	41-market story in old guide	Pipeline `**max_markets=40` ** (deliberate signal-to-parameter balance)
Data clock	Hourly mental model	`**history_fidelity=10**` (10-minute bars); walk-forward lengths **scaled** (~10d train / ~2d test per fold in 10-min steps)
Tuning	Grid emphasis	**Optuna** + pruner fix
Reporting	Figures only	**Automated Week 9 diagnostics** (attribution tables, correlation summary)

Week 8 Portfolio

#	Market	Domain	Contribution	Share	Weight
1	Will Databricks' market cap be $250B or greater at market close on IPO day?	`databricks`	0.275660	41.7%	0.0228
2	Will the Fed increase interest rates by 25+ bps after the March 2026 meeting?	`economic-policy`	0.125770	19.0%	0.0247
3	Will Giannis Antetokounmpo win the 2025–2026 NBA MVP?	`awards`	0.067983	10.3%	0.0249
4	Will Anthropic's market cap be less than $100B at market close on IPO day?	`anthropic`	0.061193	9.2%	0.0250
5	Will Chad Bianco win the California Governor Election in 2026?	`california-midterm`	0.056206	8.5%	0.0250
6	Will Stuttgart win the 2025–26 Bundesliga?	`bundesliga`	0.043709	6.6%	0.0250
7	MicroStrategy sells any Bitcoin by March 31, 2026?	`2025-predictions`	0.038356	5.8%	0.0249
8	Will Rand Paul announce a presidential run before 2027?	`celebrities`	0.026856	4.1%	0.0252
9	Will Claude 5 be released by March 31, 2026?	`claude-5`	-0.020991	-3.2%	0.0250
10	Will Meituan have the best AI model at the end of March 2026?	`gemini-3`	-0.017004	-2.6%	0.0250

Iteration Timeline

Run	N	What Changed	Sortino Δ	Diagnosis (from dev log)
Week 4	19	First prototype (Sortino-as-loss, mix=0.9)	−0.056	Unstable; crypto-heavy domain skew
Week 8	177	Category-balanced, frozen holdout	−0.149	Sortino ratio has degenerate gradients; mix=0.9 neutered optimizer
Wk9 v1	~260	Mean-downside loss, mix=0.4	−0.008	Penalties + entropy pull all weights to uniform
Wk9 v2	~260	mix=0.1, entropy=0	−0.014	260 markets = massively underdetermined
Wk9 latest	40	10-min data, Adam, Optuna	−0.009	Near-uncorrelated → equal-weight ≈ min-variance

- Each iteration identified a structural bottleneck
- Across all configurations, the optimizer converges to near-equal-weight — the return signal in short-horizon prediction market data is insufficient to overcome the penalty terms pulling toward uniform allocation

Current Results from the Sample

- Constrained portfolio is not ahead on holdout Sortino in this snapshot
- Weights in many regimes still sit near equal-weight; the project's honest punchline is iterative diagnosis (dilution, penalties, objective conditioning) and ongoing Optuna search under richer data.

Metric	Baseline	Constrained	Delta
Sortino	0.0791	0.0700	−0.0091
Max drawdown	−7.16%	−7.34%	−0.18%
Volatility	0.00336	0.00297	−0.00040

Note:

- P&L attribution is concentrated: top market/domain (databricks) ~42% of attributed return
- Risk structure: variance ratio constrained vs baseline ≈ 0.93; max abs category correlation small (~0.035 in that run).

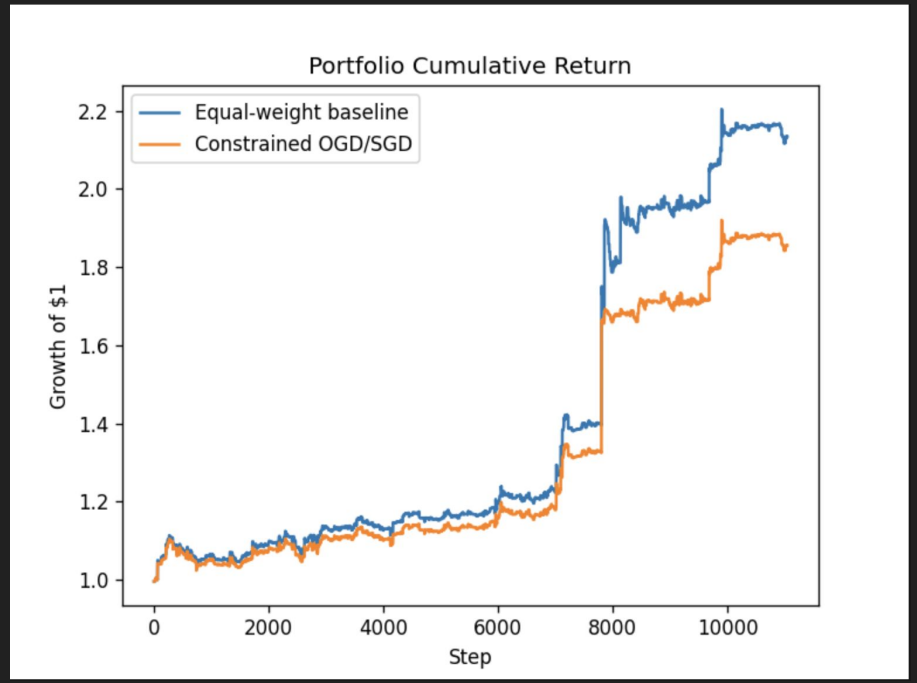
Mean Return Comparison

- Constrained model currently does not beat baseline out-of-sample on risk-adjusted return or drawdown.
- Constrained model currency also has a lower volatility than the baseline model.

Metric	Baseline	Constrained	Delta
Sortino ratio	0.0791	0.0700	-0.0091
Max drawdown	-7.1565%	-7.3414%	-0.1849%
Mean return	0.00007378	0.00005993	-0.00001385
Volatility	0.00336451	0.00296682	-0.00039769

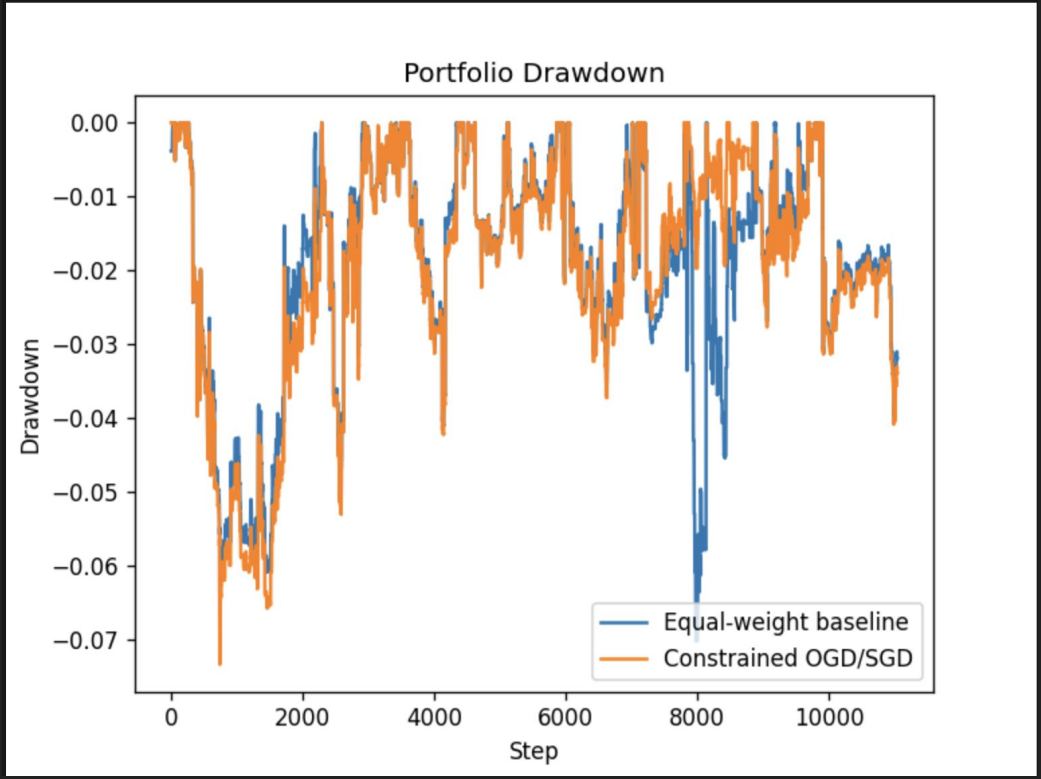
Equity Comparison

- Divergence from baseline portfolio, but it seems that our constrained portfolio actually grew slower than the equal-weighted portfolio



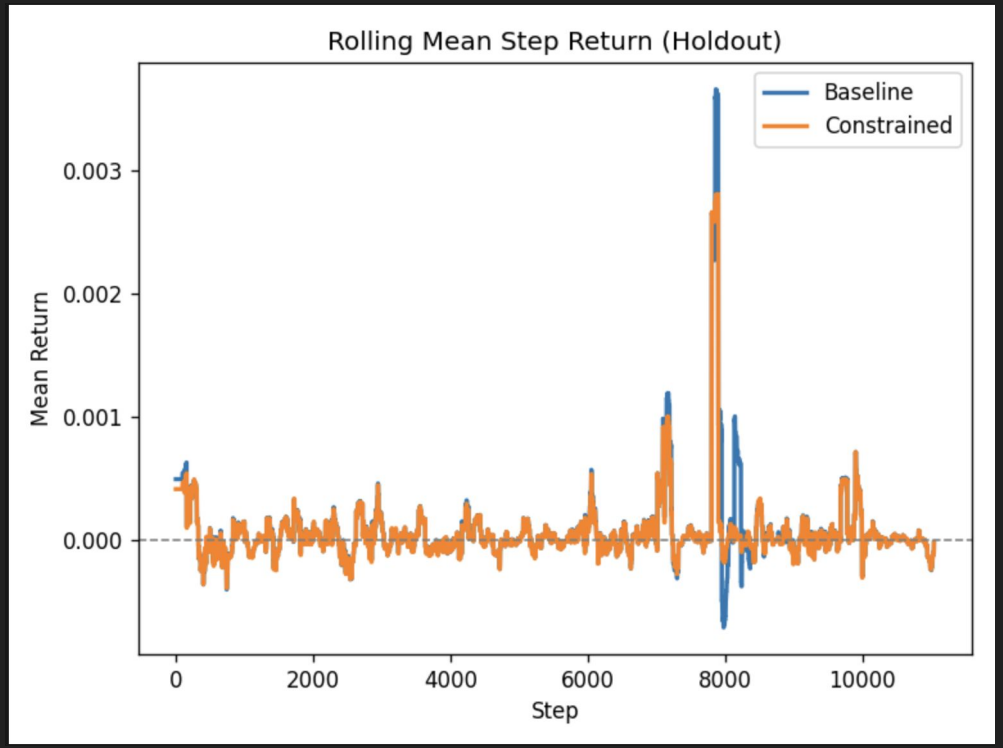
Drawdown Comparison

- Both strategies experience very similar drawdown patterns throughout the sample.
- However, our constrained portfolio avoids the largest drawdown of the baseline portfolio, which means our portfolio could provide some downside protection

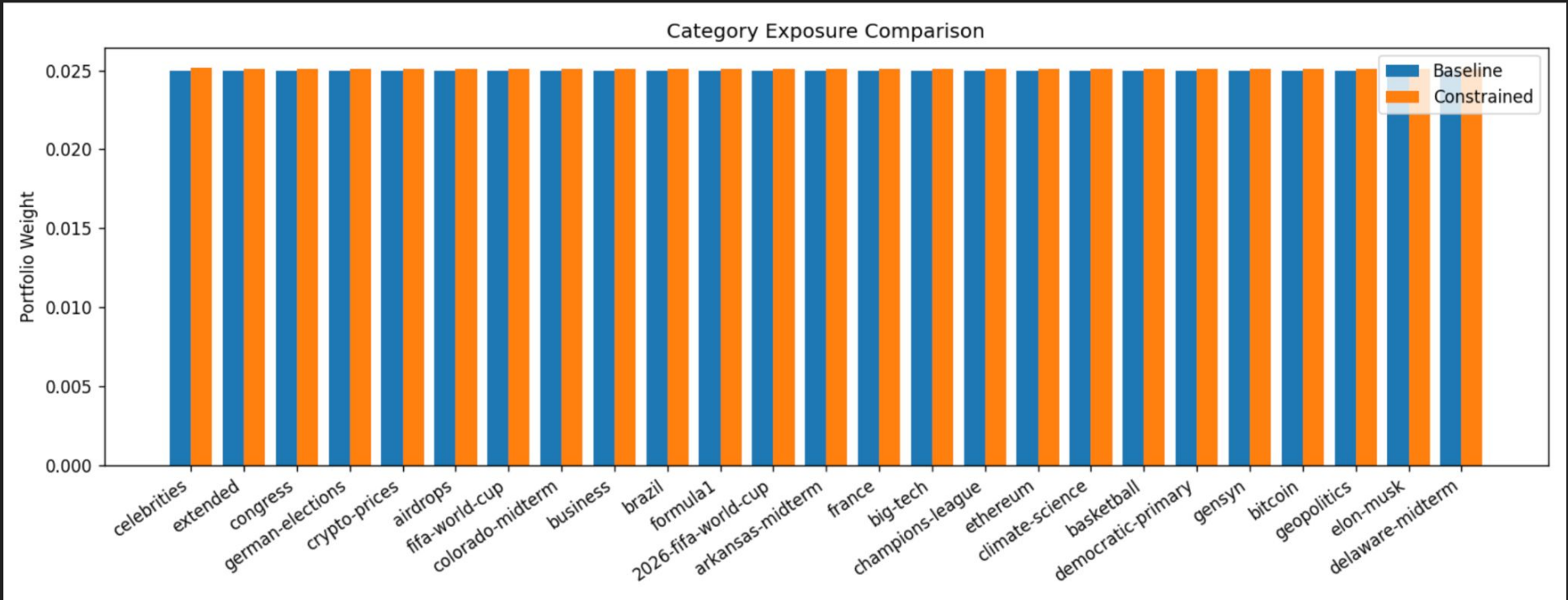


Rolling Mean Return Comparison

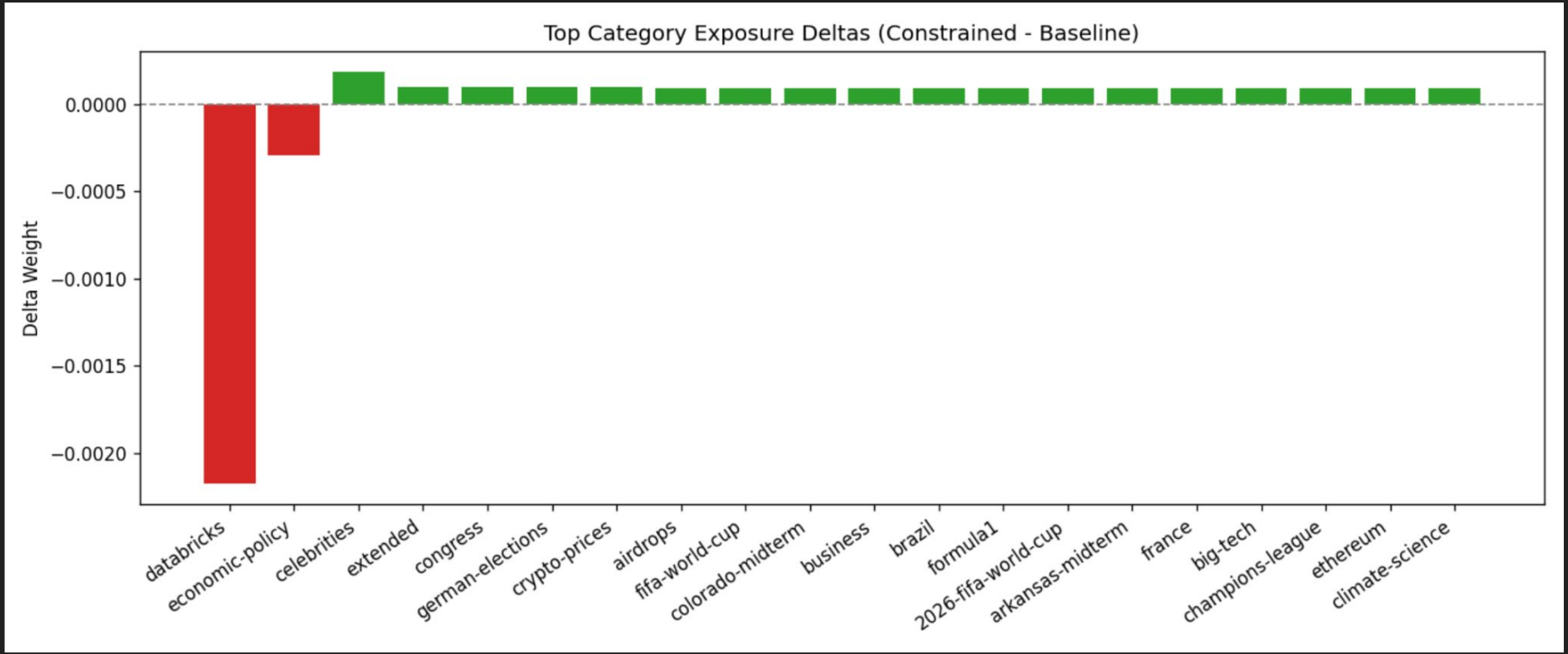
- Both strategies show highly unstable rolling mean returns that frequently cross zero.
- The constrained portfolio does not produce a more persistent positive return pattern in holdout.



Exposure Results (1/2)



Exposure Results (2/2)



Prediction Markets vs. Equity Markets

	Equity Markets	Prediction Markets (Our Setting)
Asset correlation	Correlated by sector (tech moves together, energy moves together)	Nearly independent (max category correlation: 0.035)
Return driver	Persistent, recurring factors (momentum, value)	One-off event resolution (Databricks IPO = 41.7% of total return)
Optimizer opportunity	Correlation structure provides exploitable signal	No covariance structure to exploit

- Prediction market assets are nearly independent (max category correlation: 0.035) unlike equity sectors which are correlated
- Returns are driven by one-off event resolution (Databricks IPO = 41.7%, top 2 markets = 60.7%) rather than persistent recurring factors
- No exploitable covariance structure exists for the optimizer to act on, which is why it converges to near-equal-weight

Current Progress

- Markets retained: 40
- Category count: 40
- Data / eval: 10-minute bars; online holdout; walk-forward train/test block ≈ 1440 / 288 steps (~10d / ~2d per fold in 10-min steps); ~44,168 tuning steps / ~11,042 holdout steps
- Tuning: Optuna on mean–downside objective (100 trials, 80 completed in saved week8_constrained_best_metrics.json)
- Best constrained setup (selected trial): uniform_mix ≈ 0.087, domain_limit ≈ 0.13, max_weight ≈ 0.13, rolling_window = 288, evaluation_mode = online
- Holdout baseline Sortino: 0.0791 | Holdout constrained Sortino: 0.0700 (Δ −0.0091)
- Holdout baseline max drawdown: −7.16% | Holdout constrained max drawdown: −7.34%

Next Steps

- Bootstrap CIs on Sortino: quantify whether the −0.0091 holdout delta is statistically meaningful or within noise
- Stress tests: evaluate both portfolios across different market regimes to assess robustness
- Narrower universe pre-screening: tighten market selection below 40 to improve signal-to-parameter balance
- Further Optuna trials: increase OPTUNA_N_TRIALS in the pipeline; search space already includes variance_penalties and downside_penalties tuples, so more trials may find better penalty combinations than the current 1.2 / 1.2